

Enhancing Website Search with AI

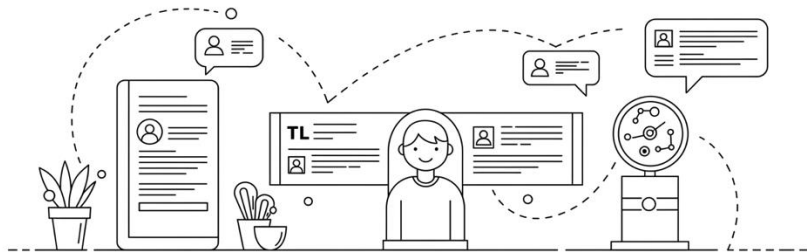
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User Experience
& Digital Strategy
Colorado College



github.com/leemark/beyond-chatbots

tl;dr



Too Long; Didn't READ

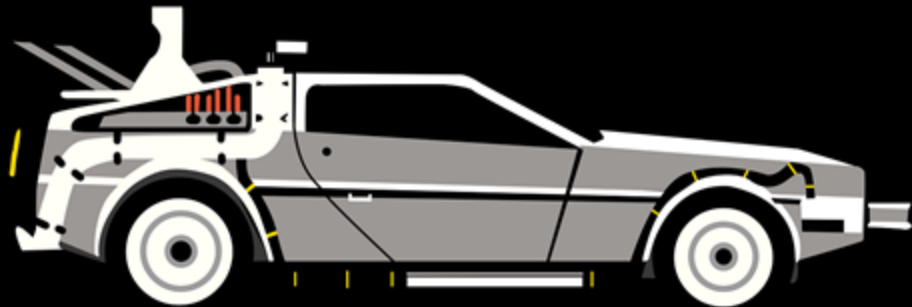
- Enhance, Don't Replace
- LLMs: A Do-Anything Multitool
- Embrace the Unexpected

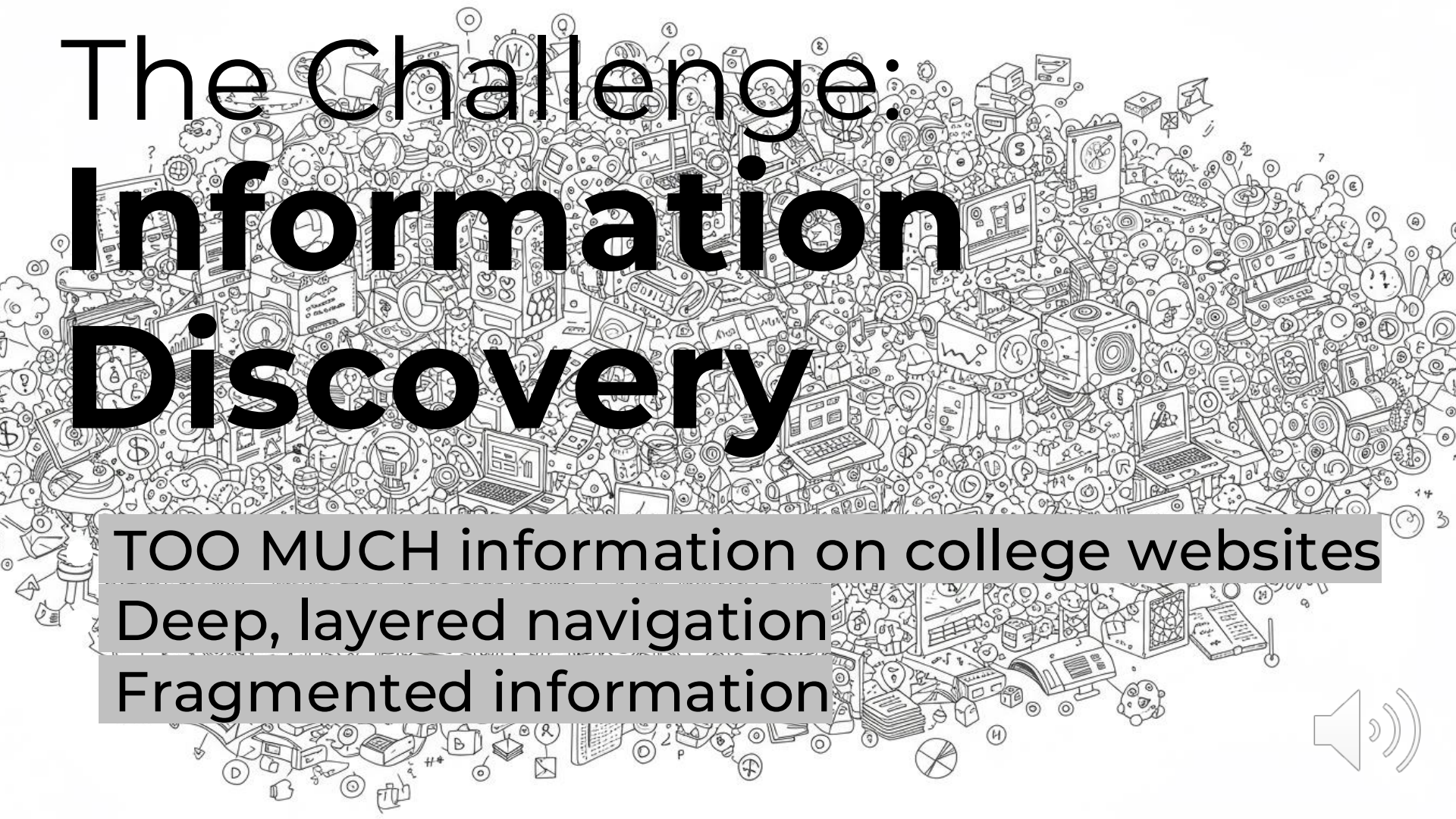


Where we're going, we don't need roads!

(but it's nice to have a roadmap anyway)

- 1) Challenges with web search
- 2) LLM chatbots and their (current) limitations
- 3) Basic to advanced LLM use
- 4) A RAG use case
- 5) The landscape ahead





The Challenge: Information Discovery

TOO MUCH information on college websites

Deep, layered navigation

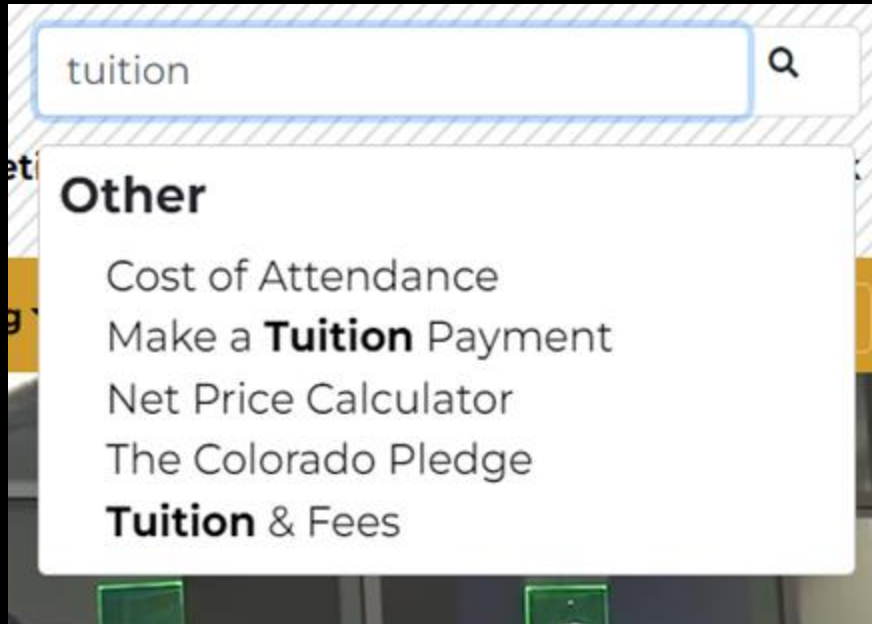
Fragmented information



Current Search Tools

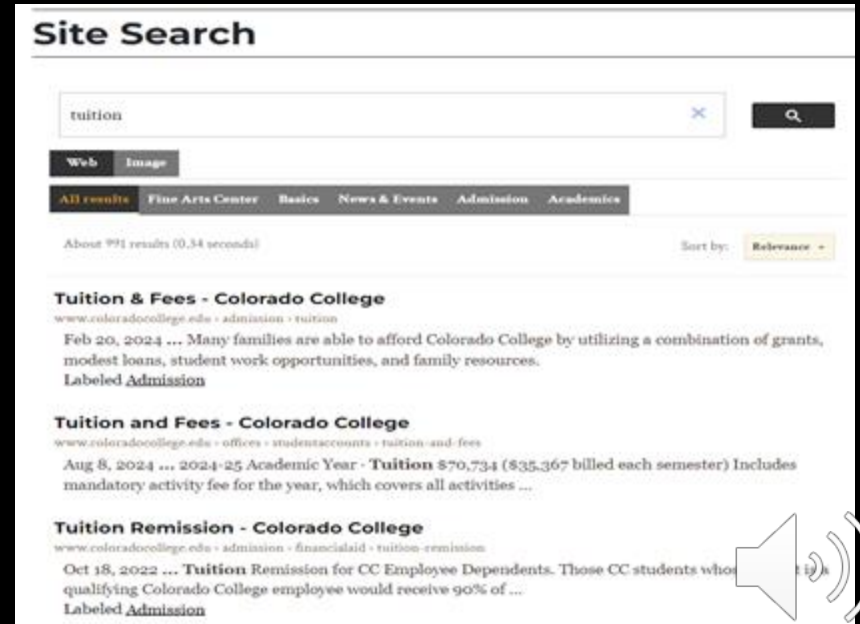
QuickSearch:

Simple keyword search



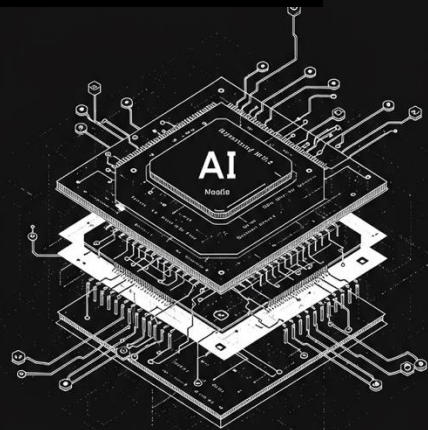
Google Custom Search:

Algorithmic; 10 blue links



Large Language Models (LLMs)

- AI that understands and generates human-like text
- Trained on massive datasets
- Uses deep learning and neural networks. Generative pre-trained transformers.



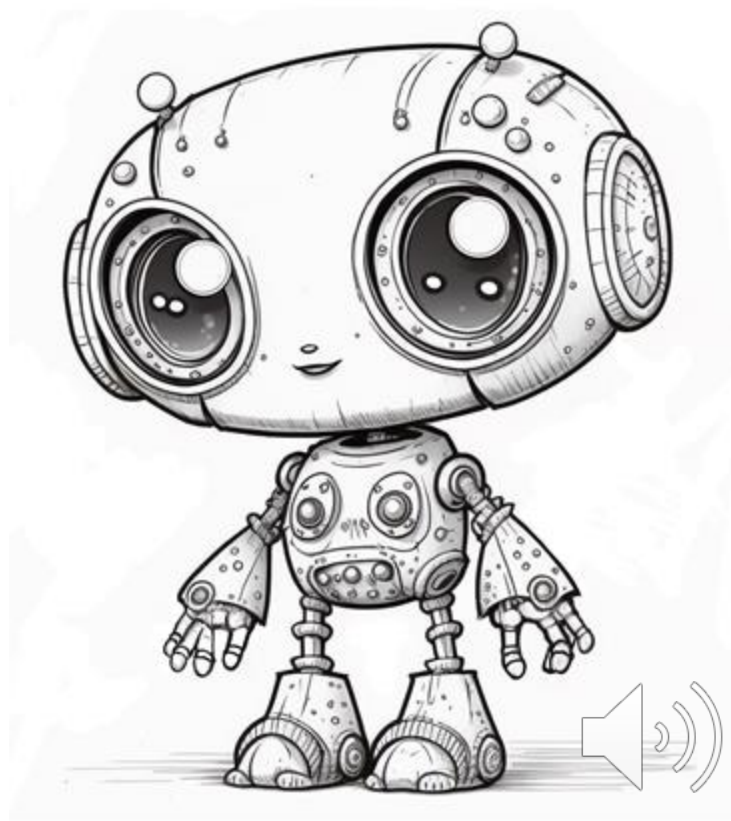
What Can LLMs Do Well?

- Translation
- Summarization
- Reasoning*
- Pattern recognition
- Question-answering
- Multimodality*



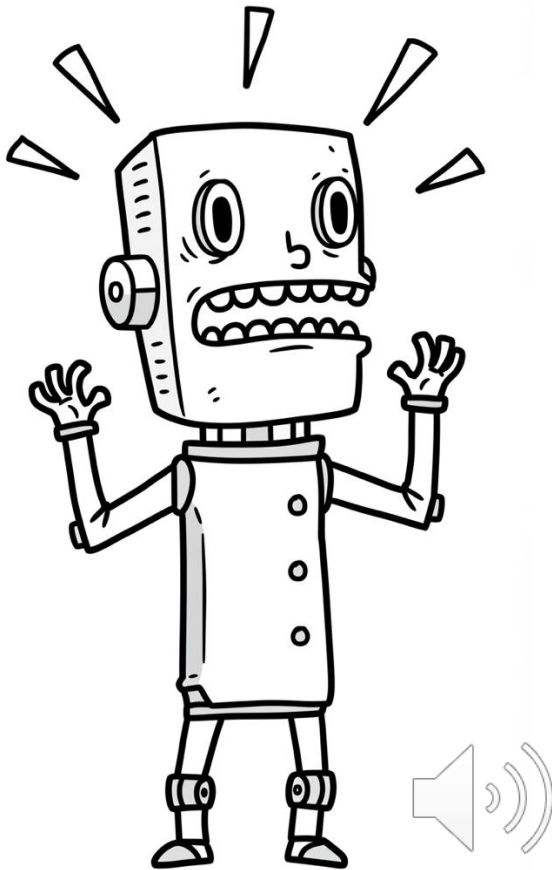
Some **Advantages** of LLM Chatbots

- Natural language queries
- Diverse language tasks
- Human-like responses
- Continuous improvement



Some **Limitations** of LLM Chatbots

- Potential for "hallucinations"
- Bias in training data
- Security and privacy concerns
- Limited to training data cutoff
- Limited context window (sometimes)



LLMs in Action

How much is tuition at Colorado College?

You're giving feedback on a new version of ChatGPT.

Which response do you prefer? Responses may take a moment to load.



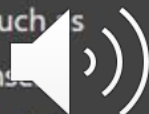
Response 1

For the 2023-2024 academic year, the tuition at Colorado College is \$68,634. This amount does not include additional costs such as room and board, fees, books, and personal expenses, which can add several thousand dollars to the total cost of attendance.



Response 2

As of the 2023-2024 academic year, the tuition at Colorado College is \$70,692 per year for undergraduate students. This does not include additional costs such as room and board, fees, books, and personal expenses, which can bring the total annual cost to approximately \$91,672.



how much is tuition



Web

Image

All results

News & Events

Blogs

Academics

Admission

Basics

About 215 results (0.19 seconds)

Sort by:

Relevance ▾

Tuition & Fees - Colorado College

www.coloradocollege.edu > admission > tuition

Feb 20, 2024 ... **Many** families are able to afford Colorado College by utilizing a combination of grants, modest loans, student work opportunities, and family resources. Labeled [Admission](#)

Tuition and Fees - Colorado College

www.coloradocollege.edu > offices > studentaccounts > tuition-and-fees

Aug 8, 2024 ... 2024-25 Academic Year · **Tuition** \$70,734 (\$35,367 billed each semester) Includes mandatory activity fee for the year, which covers all activities ...

Tuition and Fees - Colorado College

www.coloradocollege.edu > offices > CC-students > tuition_and_fees

Apr 29, 2024 ... Block C courses will be entirely remote, and limited student housing will be available. Housing and meal plans rates are subject to change.

Cost of Attendance - Colorado College

www.coloradocollege.edu > admission > handbook > cost-of-attendance

Feb 20, 2024 ... CC Financial Aid & You: Partnership for Success. **Many** families are able to afford Colorado College by utilizing a combination of grants, modest ... Labeled [Admission](#)

Tuition and Fees - Colorado College

www.coloradocollege.edu > pre-college > precollegetuitionindex

Need-Based Scholarship. As part of your online course application, you will need to submit a 500 word essay on "The Importance of a Liberal Arts Education in ...



How much is tuition at Colorado College?

🔍 Sources

Colorado College - Net Price, Tuition, Cost to Attend, Financial Aid ...

 collegesimply · 1

The Colorado College - College Confidential

 collegeconfide... · 2

Tuition & Fees - Colorado College

 coloradocollege · 3



View 5 more

✨ Answer

The tuition at Colorado College for the 2024-25 academic year is \$70,224 ³ ⁵. This amount covers full-time enrollment for the entire academic year, which consists of two semesters.

Some key points about Colorado College's tuition and costs:

- The \$70,224 tuition includes a mandatory student activity fee of \$510 ³.
- Tuition is the same for both in-state and out-of-state students, as Colorado College is a private institution ².
- The total estimated cost of attendance for 2024-25, including tuition, fees, room and board, books, and other expenses, is \$91,154 ³.



Ask follow-up



Pro



perplexity.ai

Basic **prompting**

The art of crafting input text that elicits a specific response or behavior from the LLM.

Three key elements make a good prompt:

- **clarity** (use simple language)
- **specificity** (define exactly what you want)
- **context** (provide relevant background)



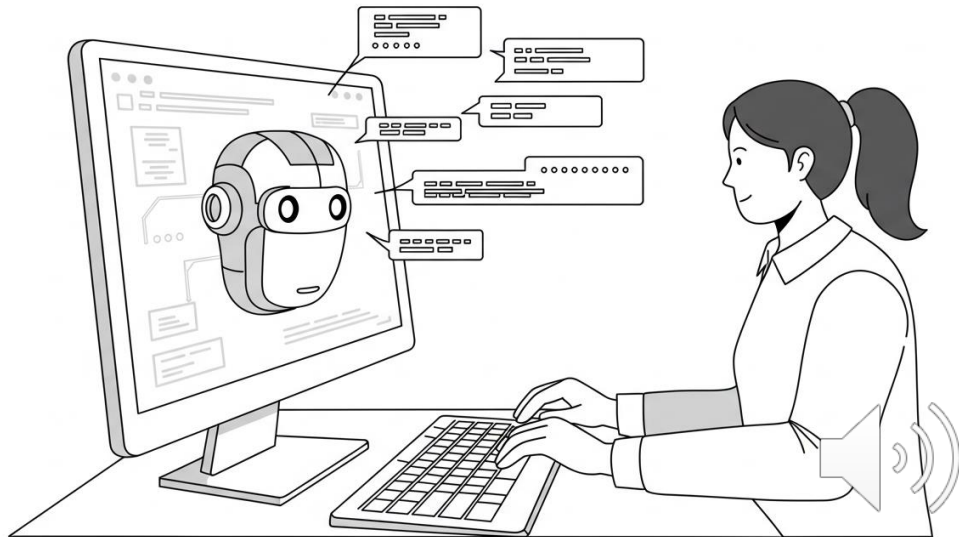
Further **Improving** LLM **Performance**

- Better prompting techniques
- "Think out loud" approach, let the LLM think
- Leverage speed and context windows
- Use the right tool for the job
- Let's start to think about how we can build a framework around the LLM to add guardrails



Advanced Prompting Techniques

- Chain-of-thought prompting
- Few-shot or many-shot (in-context) learning
- Pre-fill/partial response
- <scratchpad>
- Reflection



Chain-of-Thought Prompting

- Encourages step-by-step reasoning
- Improves accuracy for complex tasks
- Example:
"Think through this problem step-by-step..."



Prompting Techniques: From Zero to Many

Zero-shot = No examples provided

*Classify the sentiment of the following sentence:
'The product exceeded my expectations.'*

Few-shot = 1-5 examples

Classify the sentiment of the following sentences as positive, neutral, or negative:

- 1. 'The customer service was terrible.' -> Negative*
- 2. 'The meal was okay, nothing special.' -> Neutral*
- 3. 'The product exceeded my expectations.' ->*

Many-shot = 10+ examples



Pre-filled Response

- Start the response for the AI
- Guides the structure and content
- Example:

*The three main dining locations on campus are:
(1) _____*



The <scratchpad> Technique

Ask the AI to show its work, for example in <scratchpad> XML tags, before composing its answer. Improves transparency and accuracy.

```
<scratchpad>
```

```
Key points supporting AI achieving consciousness:
```

1. Increasing complexity of AI systems mimicking brain structure
2. Advancements in neural network architectures
3. Growing computational power approaching brain-level processing
4. Emergence theory - consciousness as an emergent property of complex systems
5. Gradual development of self-awareness in AI systems
6. Potential for AI to model its own thought processes
7. Information Integration Theory of consciousness
8. Functionalist arguments - consciousness arising from functional organization
9. Evolutionary perspective on consciousness development
10. Technological singularity leading to super intelligent AI
11. Quantum computing potentially enabling consciousness-like states
12. Philosophical arguments for multiple realizability of mental states
13. Advancements in understanding and replicating neurobiological processes
14. Potential for AI to develop subjective experiences or qualia
15. Machine learning systems showing signs of creativity and intuition

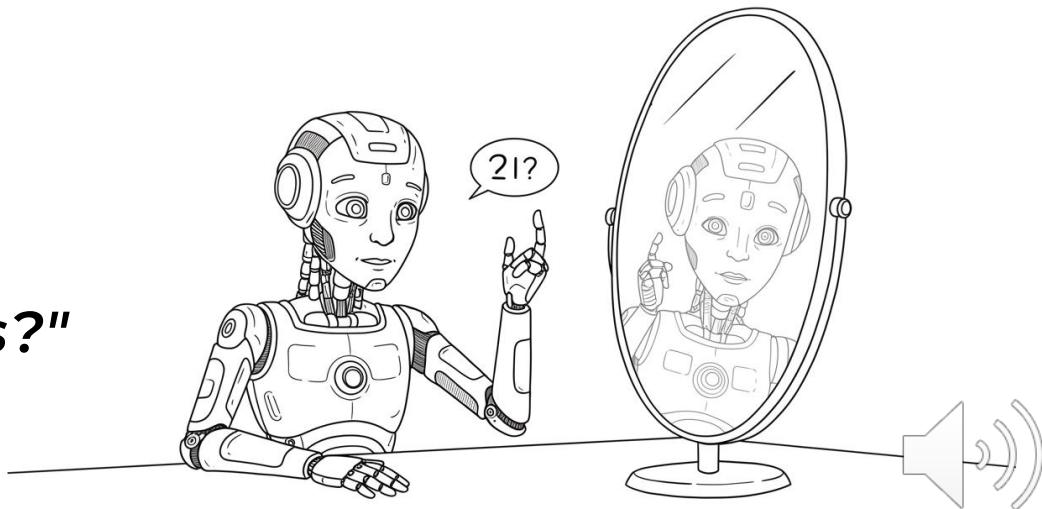
```
</scratchpad>
```



Reflection: Self-Improvement for AI

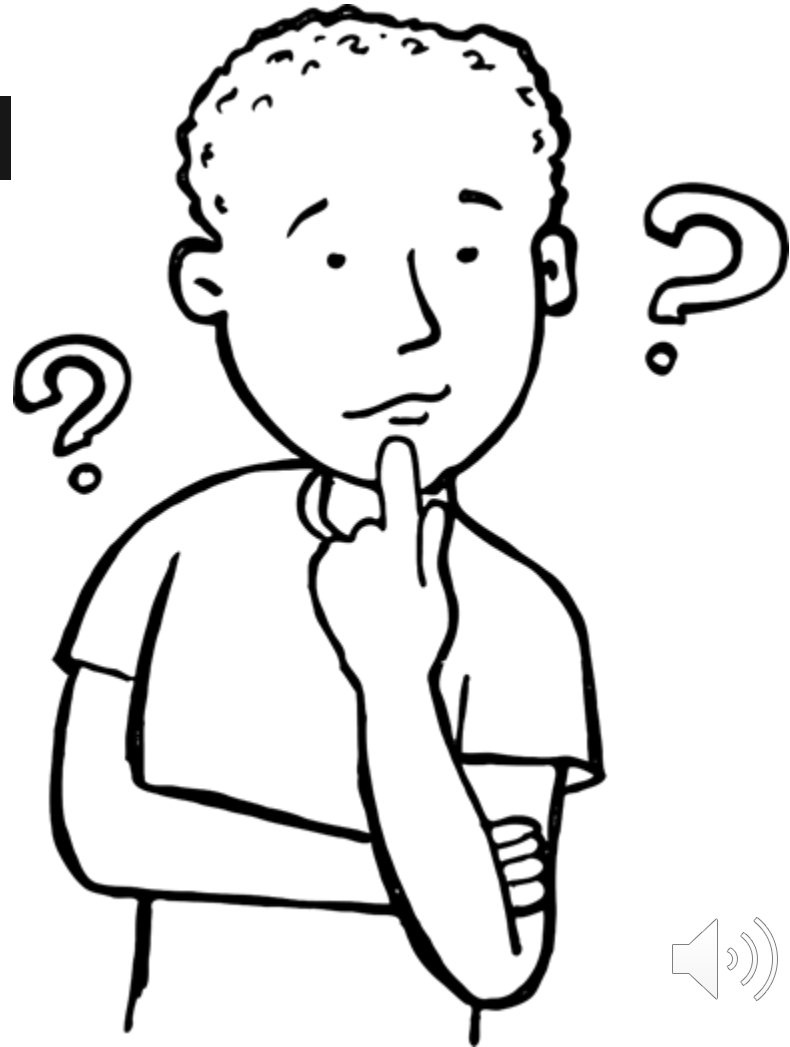
The LLM reviews & improves
its own response

Example:
*"Review your answer.
Can it be improved?
Do you see any mistakes?"*



Choosing a model

- Open vs closed
- Cost vs intelligence
- Cost and speed
- Intelligence vs speed

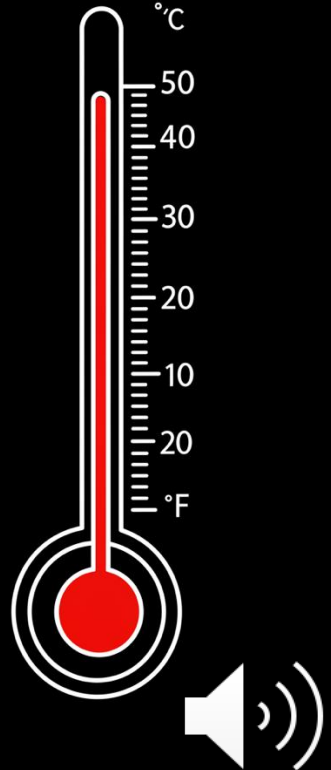


Turning up the **Temperature**

Temperature Controls Randomness

Higher values
= more creative, varied responses

Lower values
= more focused, deterministic outputs



System prompts

“You are
a helpful
assistant...”



A proposed methodology for better Web search

R.A.G.

Retrieval-Augmented Generation



Why Use RAG?

- Contextual, natural language answers
- References and citations
- Follow-up questions for deepened engagement
- Combines LLM capabilities with up-to-date info



Can I study Microbiology at your school?

IDK, you're gonna have to look for yourself... you can try
<https://example.edu/science/biology/microbiology>

or
<https://example.edu/departments/microbiology/research-projects>

or maybe
<https://example.edu/faculty/microbiology-lab-resources>

or possibly
<https://example.edu/cour>



Text Message



Can I study Microbiology at your school?

IDK for sure, I don't know everything I should know about this school

but my system prompt tells me to be a helpful assistant and my RLHF has trained me to be a people pleaser...

so here is a bunch of stuff I just made up about studying microbiology here

p.s. please like me 🥺



Text Message



Can I study Microbiology at your school?

Absolutely you can, here is some factual information about studying microbiology at this school

and a short list of some of the actual courses offered

it's sourced from the .edu website, and here is a link to where I found the information

let me know if you need anything else 🙌



Text Message



Ask Colorado College

Enter your question:

Where do first year students live on campus?

Search

Answer: First-year students at Colorado College are housed in the "Big 3" traditional halls: Mathias Hall, Loomis Hall, and South Hall [1].

Related questions:

What are the amenities like in the Big 3 residence halls?

Are there single rooms available in the first-year residence halls?

What is the social atmosphere like in the first-year dorms?

Sources used:

[1] <https://www.coloradocollege.edu/offices/housing-and-conferences/livingatcc.page/first-year.page.html>

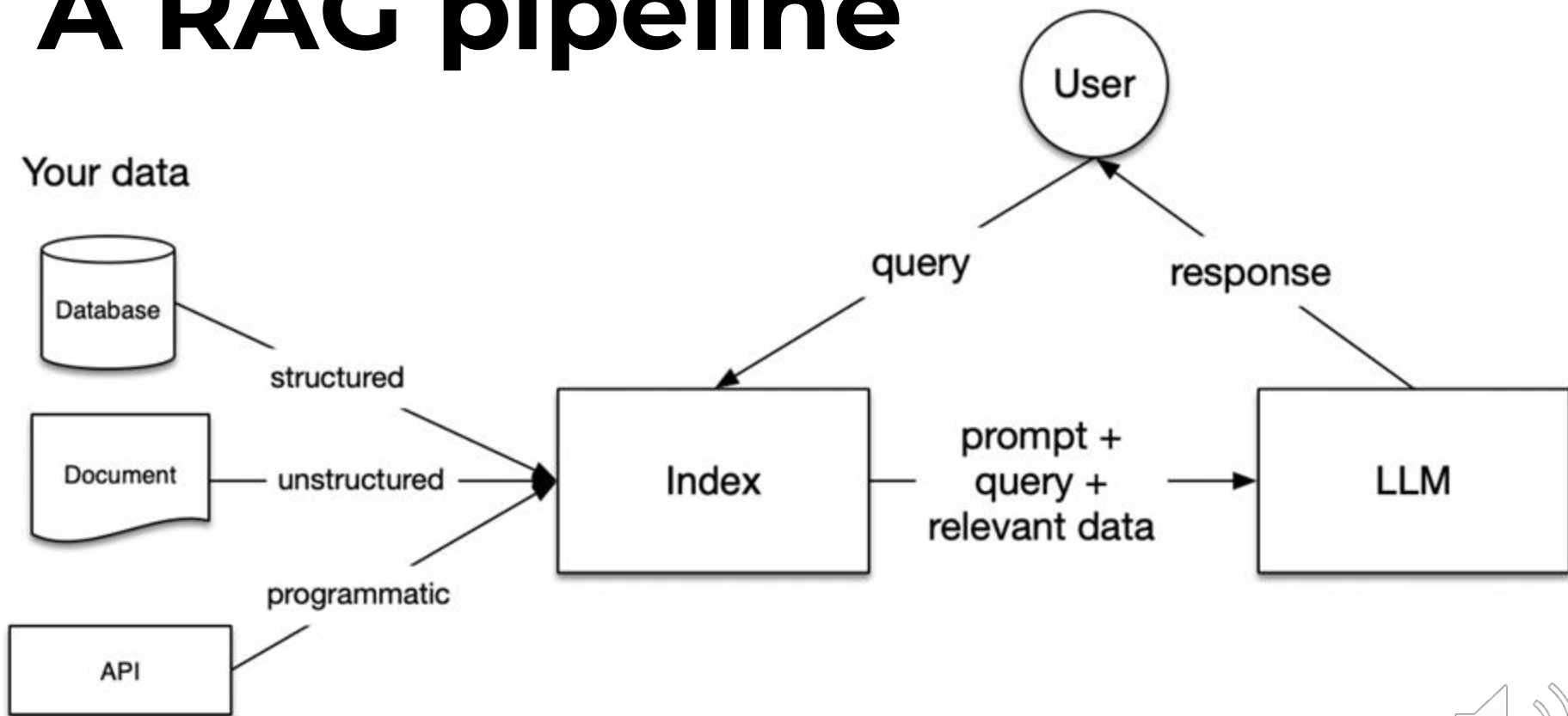


it's **NOT** replacing what would normally be a human-to-human interaction with a bot

it's replacing what is already a human-to-computer interaction with a **better** human-to-computer interaction



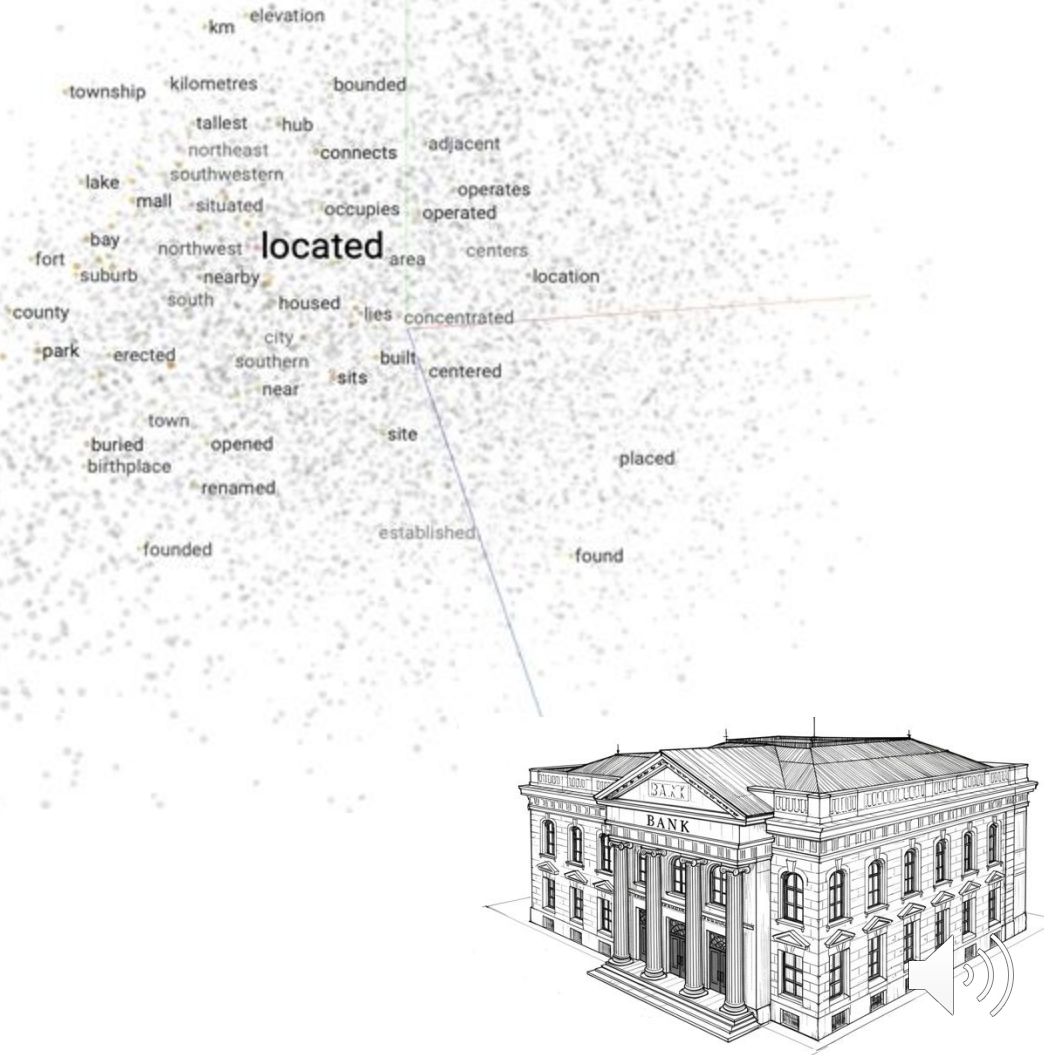
A RAG pipeline



Use an LLM to evaluate & rewrite the user query (1 or n variations)
Search information sources (search API: Google, Brave, DDG)
Grab and parse the contents of resultant pages
Store results in vector database (optionally*, for semantic search)
Retrieve relevant info (optional, for semantic search)
Pass original user query plus relevant context to LLM to generate the answer (don't forget to use our advanced prompting techniques)
Use LLM to suggest follow-up questions
Send all to user: generated answer, citations, and follow-up questions



- Embed content into high-dimensional space
- Semantic similarity search
- Efficient retrieval of relevant information



```
def main():
```

```
    # Step 1: Rephrase the user's query
```

```
    rephrased_query = rephrase_query(USER_PROMPT)
```

```
    print(f"Rephrased Query: {rephrased_query}")
```

```
    # Step 2: Search for relevant URLs
```

```
    urls = search_urls(rephrased_query)
```

```
    # Step 3: Scrape and parse content from the URLs
```

```
    contents, sources = scrape_and_parse(urls)
```

```
    # Step 4: Embed the scraped content into the Chroma database
```

```
    embed_into_chroma_db(contents, sources, "colorado_college_tuition")
```

```
    # Step 5: Query the Chroma database for relevant information
```

```
    relevant_docs, relevant_sources = query_chroma_db(rephrased_query, "colorado_college_tuition")
```

```
    # Step 6: Prepare the context for the response generation
```

```
    context = "\n".join([f"Source: {source}\nContent: {doc}" for doc, source in zip(relevant_docs, relevant_sources)])
```

```
    # Step 7: Generate the final response
```

```
    response = generate_response(context, rephrased_query)
```

```
    # Step 8: Print the results
```

```
    print("Response:")
```

```
    print(response)
```

```
    print("\nRelevant Sources:")
```

```
    print("\n".join(relevant_sources))
```

How to build rudimentary **RAG-based search** in Python



Rephrasing user queries

```
{"role": "system", "content": "You are a search  
assistant. Rephrase the following as a concise  
search query expertly crafted to retrieve  
relevant results from a search engine. Return  
only the query, no explanation:"},
```

```
{"role": "user", "content": user_query_goes_here}
```



Searching for relevant URLs

```
url = f"https://api.search.brave.com/res/v1/web/search?q={query}"
```

```
headers = {"X-Subscription-Token": BRAVE_API_KEY}
```

```
response = requests.get(url, headers=headers)
```

```
results = response.json()
```



Scraping & parsing content

```
response = requests.get(url, headers=headers)
```

```
soup = BeautifulSoup(response.content, 'html.parser')
```

```
contents.append(soup.get_text())
```

```
sources.append(url)
```



Embedding content into a vector database

```
client = chromadb.PersistentClient()
collection = client.get_or_create_collection(collection_name)
for content, source in zip(contents, sources):
    collection.add(
        documents=[content],
        metadatas=[{"source": source}],
        ids=[str(uuid.uuid4())]
    )
```



Querying the database for relevant information

```
results = collection.query(  
    query_texts=[query],  
    n_results=top_k,  
    include=["documents", "metadatas"]  
)
```



Generating the response

prompt = f"""You are a helpful assistant that answers questions about Colorado College **only based on information from the CC website**, provided below.

Question: {question}

Context: {context}

Instructions:

1. If you have enough information to answer the question confidently and accurately, provide a direct answer.
2. **If you don't have enough information** to answer the question appropriately then reply with a brief ...



Ask Colorado College

Enter your question:

Where do first year students live on campus?

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Answer: First-year students at Colorado College are housed in the "Big 3" traditional halls: Mathias Hall, Loomis Hall, and South Hall [1].

Related questions:

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Are there single rooms available in the first-year residence halls?

What is the social atmosphere like in the first-year dorms?

Sources used:

[1] <https://www.coloradocollege.edu/offices/housing-and-conferences/livingatcc.page/first-year.page.html>



Thoughts after building this.

The first 80% is easy, the last 5% of testing and refinement is where the real challenge and the real learning is.

Beyond search, it hints at new, methods for web content Quality Assurance.

Consider *also* having a very well-vetted information source in the pipeline/index.

Enter your question:

How much is tuition?

Search



Answer: Tuition for the 2024-25 academic year is \$62,070. [1]

Tuition and Fees:	\$62,070.00
Books and Supplies:	\$1,240.00
Food:	\$1,188.00



Ask Colorado College

Enter your question:

¿Dónde puedo comer en el campus?

Search

Answer: En el campus, puedes comer en Rastall Dining Hall o Benjamin's, ambos ubicados en el Worner Campus Center. [1] Rastall ofrece un buffet con una variedad de opciones, incluyendo comida tradicional, vegana y vegetariana, mientras que Benjamin's tiene una parrilla con hamburguesas y la opción de crear tu propio plato de fideos.

Related questions:

¿Qué tipo de comida se sirve en Rastall Dining Hall?

¿Hay opciones sin gluten disponibles en el campus?



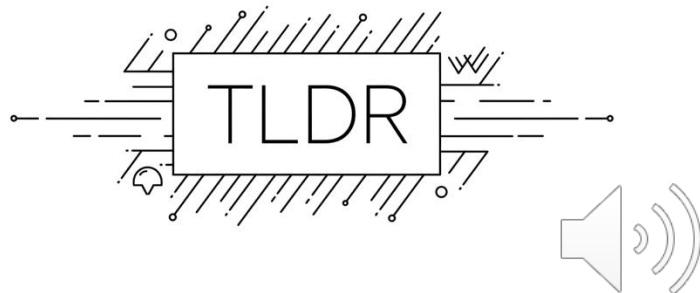
The Future (or present?) of AI-Powered Web Experiences

- Multimodal search (text, image, audio)
- Personalization
- Conversational search
- Integration with other campus systems
- Autonomous Agents



tl;dr revisited

- Enhance, Don't Replace
- LLMs: Your AI Multitool
- Embrace the Unexpected



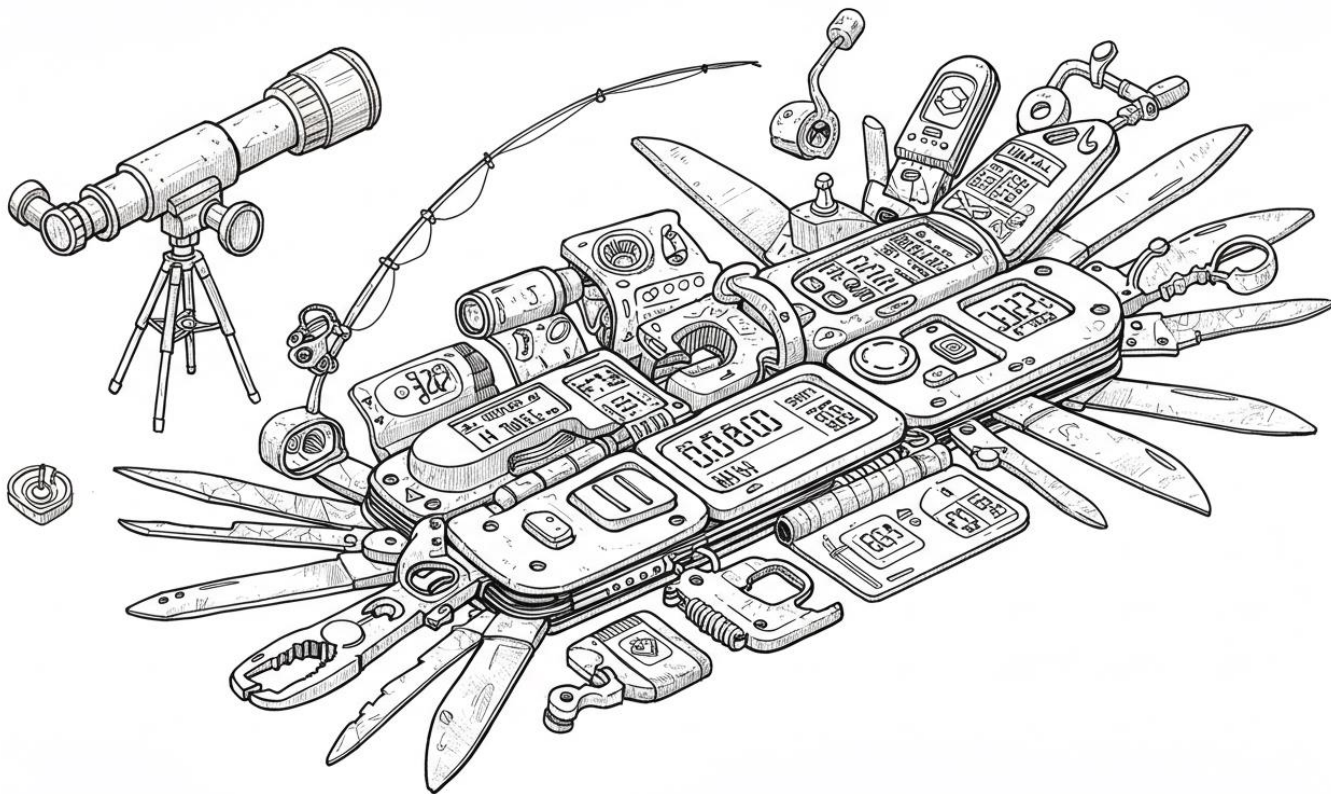
Enhance, Don't Replace

choose

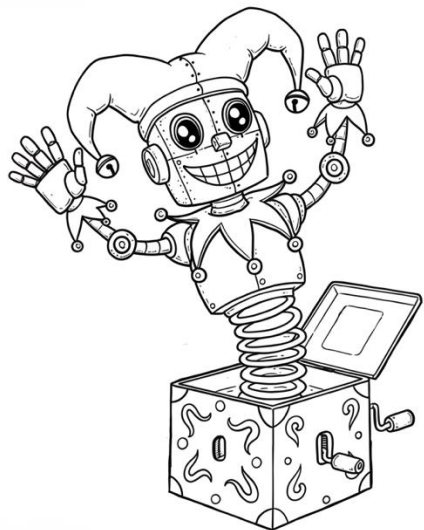
Evolution over **Revolution**



A **Do-Anything** Multitool



Embrace the Unexpected



A few **final** thoughts

- LLMs CAN offer powerful information search capabilities
 - But ONLY with “fact checking” and accurate info
- RAG combines the power of LLMs with up-to-date info from traditional search
- Keep learning, keep exploring.
Things are changing fast.



Thank You!

Questions?

Resources,
code examples,
slides, etc.

github.com/leemark/beyond-chatbots

